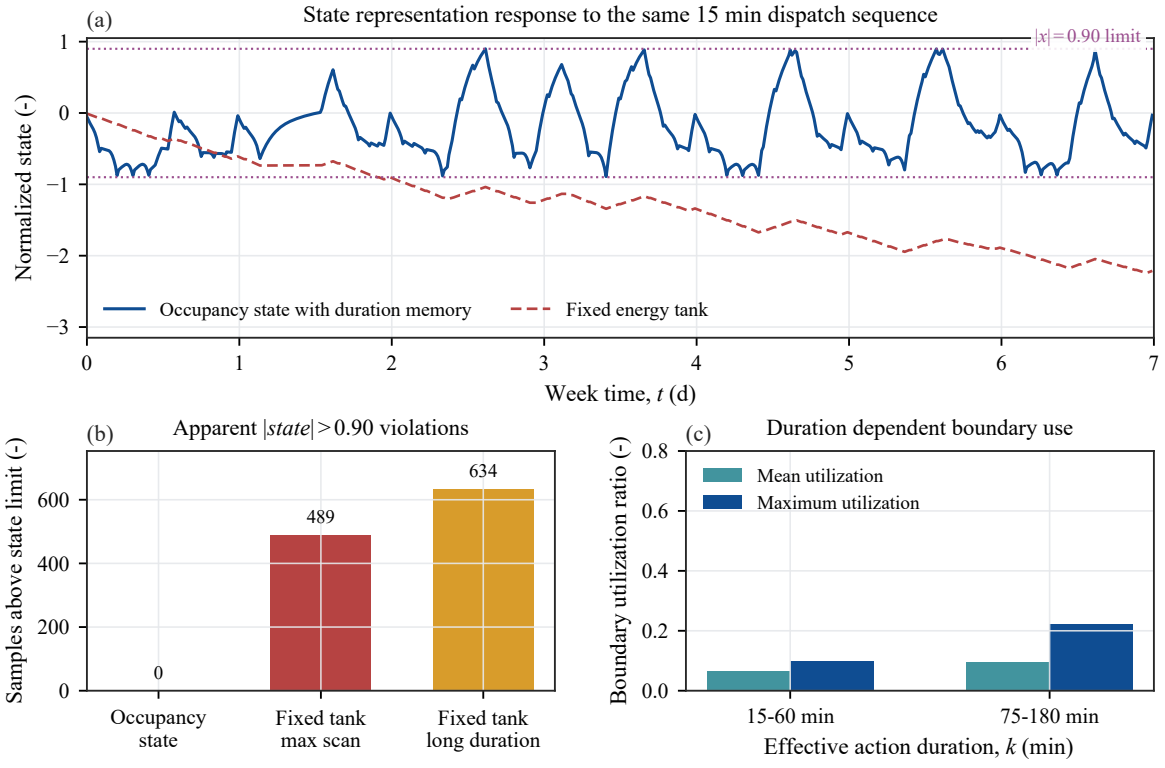

(a)
State representation response to the same 15 min dispatch sequence
|x| = 0.90 limit
Normalized state (-)
1
0
−1
−2
−3
Occupancy state with duration memory
Fixed energy tank
0
1
2
3
4
5
6
7
Week time, t (d)
(b)
Apparent |state| > 0.90 violations
Samples above state limit (-)
600
400
200
0
489
634
0
Occupancy state
Fixed tank max scan
Fixed tank long duration
(c)
Duration dependent boundary use
Boundary utilization ratio (-)
0.8
0.6
0.4
0.2
0.0
Mean utilization
Maximum utilization
15-60 min
75-180 min
Effective action duration, k (min)